Euclid's Elements — Book I
Proposition 14

Formal Reconstruction — Technical Documentation

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Converse straight-line criterion from adjacent angles

1 Introduction

Euclid's Proposition I.14 is the converse in spirit of the preceding Proposition I.13.

Proposition I.13 starts from a straight-line configuration and proves that the adjacent angles are together equal to two right angles. Proposition I.14 reverses this direction: if two adjacent angles are together equal to two right angles, and their outer rays lie on opposite sides of the common straight line, then those outer rays form one straight line.

In Euclid's original proof the argument is indirect. The ray opposite to one of the given rays is constructed, Proposition I.13 is applied, and equality of the resulting angles forces the constructed ray to coincide with the remaining given ray.

The Hilbert reconstruction follows the same geometric idea at a more structural level. The condition "equal to two right angles" is encoded without numerical angle measure, using an opposite ray and angle congruence. The decisive step is then supplied by the uniqueness clause of Hilbert's angle-construction axiom.

Proposition I.14 is therefore an instructive example of the interaction between Euclid's angle language, Hilbert plane separation, angle construction, and the ray calculus developed in Book Zero.

2 The Classical Configuration

Let AB be a straight line, and let two further straight lines BC and BD meet it at the common point B, with C and D lying on opposite sides of AB.

The two adjacent angles are therefore

$$\angle ABC \qquad \text{and} \qquad \angle ABD.$$

The hypothesis of Proposition I.14 is that these two angles are together equal to two right angles. The conclusion is that the two outer rays BC and BD are opposite rays and hence form one straight line.

3 Statement

Theorem 1 (Euclid I.14)**.** *Let AB be a straight line, and let the straight lines BC and BD meet it at the point B on opposite sides of AB.*

If the adjacent angles

$$\angle ABC \qquad \textit{and} \qquad \angle ABD$$

are together equal to two right angles, then BC and BD form one straight line.

Equivalently,

$$C - B - D.$$

The proposition may therefore be viewed as a converse to the straight-line angle relation established in Proposition I.13.

The geometric content of the conclusion is stronger than mere collinearity. The point B lies between C and D, so the rays BC and BD are opposite rays with common endpoint B.

4 Classical Proof

Euclid argues by extending one of the two outer rays through the common vertex; the auxiliary configuration is shown in Figure 1.

Extend CB through B to a point E. Then

$$C - B - E,$$

so BC and BE form one straight line.

By Proposition I.13, the adjacent angles

$$\angle ABC \qquad \text{and} \qquad \angle ABE$$

are together equal to two right angles.

But by hypothesis,

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are also together equal to two right angles.

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

But if BE and BD were distinct, one of these two angles would be a proper part of the other. A proper part cannot be equal to the whole. Hence the rays BE and BD cannot be distinct.

Therefore BE coincides with BD. Since

$$C - B - E,$$

it follows that

$$C - B - D.$$

Thus BC and BD form one straight line.

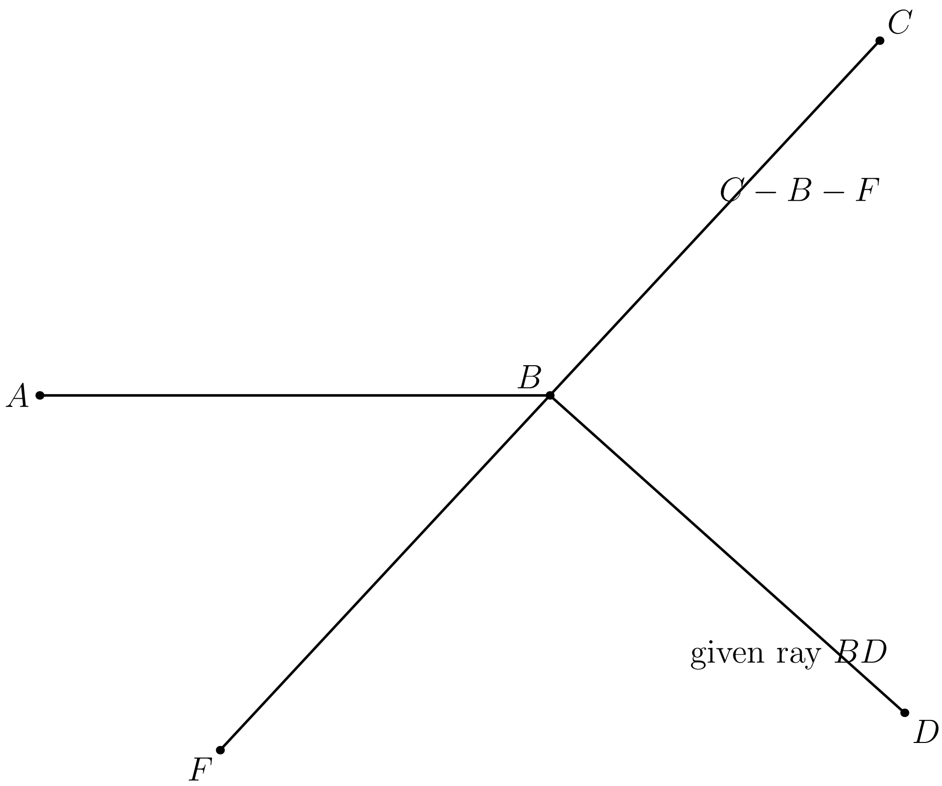


Figure 1: Classical Wilkins pre-coincidence configuration. The straight line CBF is the constructed extension through B, while BD is the given ray. The point F is Wilkins's classical auxiliary point; the formal Lean witness for the same supplementary pattern is called E. The drawing does not assume that BF and BD already coincide.

The essential structure of Euclid's argument is therefore

$$\text{extension} \longrightarrow \text{I.13} \longrightarrow \text{equality of the remaining angles} \longrightarrow \text{coincidence of the rays.}$$

The classical auxiliary diagram is genuinely useful, but it must be read as a *pre-coincidence* configuration. It would be misleading to draw BD as a visibly different ray and simultaneously treat $\angle ABF = \angle ABD$ as already geometrically realized. The equality is precisely what forces the two rays to coincide. Wilkins makes the missing uniqueness step explicit by using the fact that D and F lie on the same side of AB.

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction preserves the central geometric idea of Euclid's proof but reorganizes its logical dependencies.

Euclid extends CB through B to a point E and uses Proposition I.13 to obtain the supplementary angle relation. He then compares this relation with the hypothesis of Proposition I.14 and concludes that the rays BD and BE must coincide.

In the formal reconstruction, the point E is incorporated directly into the synthetic representation of the hypothesis. We require

$$C - B - E$$

together with

$$\angle ABD \cong \angle ABE.$$

Thus the intermediate appeal to Proposition I.13 is no longer needed: the supplementary configuration required by Euclid's argument is already present in the formal hypothesis.

The remaining problem is to prove that BD and BE determine the same ray from B.

The current source also derives

$$\mathrm{OppositeSide}(C, E; \mathrm{base})$$

from $C - B - E$ and the fact that B lies on the base line. This is a valid geometric consequence, but in the present production proof the local fact `hOppCE` is not used after it is established. It is therefore not a dependency edge of the subsequent argument and does not receive a separate figure.

At this point the reconstruction divides into two cases according to whether C, D, E are collinear.

```
Hilbert hypothesis
------------------
C-B-E
angle ABD ~= angle ABE
C and D opposite sides of base AB
        |
        v
derive: C and E opposite sides of base AB
        |
        v
case split: are C,D,E collinear?
        |
   +----+----+
```

```
   |         |
  yes        no
   |         |
line         D and E are on the same side of AB
uniqueness   |
   |         v
   |     Hilbert III.4 angle uniqueness
   |         |
   |     BE and BD are the same ray
   |         |
   +----+----+
        |
transport C-B-E along the identified ray
        |
        v
C-B-D
```

The audited presentation keeps only the auxiliary classical configuration. The earlier stand-alone statement picture added no new geometric state: the same rays and base line are already present in the extension diagram. The formal proof, however, contains a genuine geometric split according to whether C, D, E are collinear. Those two states are recorded together in Figure 2.

In the collinear case, incidence and line uniqueness are sufficient.

In the noncollinear case, plane separation first places D and E on the same side of the base line. The uniqueness clause of Hilbert's angle-construction axiom then identifies the rays BD and BE. Ray calculus and transport of betweenness finally yield

$$C - B - D.$$

Schematically, the two proof architectures are

$$\begin{aligned}
\text{Euclid} \quad &: \quad \text{extension} \to \text{I.13} \to \text{angle comparison} \\
& \to \text{ray coincidence}, \\
\text{Hilbert reconstruction} \quad &: \quad \text{encoded supplement} \to \text{case split} \\
& \to \text{line/angle uniqueness} \to \text{betweenness}.
\end{aligned}$$

6 Hilbert Reconstruction

The Hilbert reconstruction begins by replacing Euclid's phrase "together equal to two right angles" by a purely synthetic configuration.

No numerical measure of angles and no operation of angle addition are introduced. Instead, the supplementary relation is represented by an opposite ray together with angle congruence.

6.1 Encoding the Angle Hypothesis

Choose a point E on the continuation of CB beyond B, so that

$$C - B - E.$$

The rays BC and BE are therefore opposite. Consequently, $\angle ABE$ is the angle supplementary to $\angle ABC$.

The hypothesis that

$$\angle ABC \quad \text{and} \quad \angle ABD$$

are together equal to two right angles is represented by requiring

$$\angle ABD \cong \angle ABE.$$

The same supplementary pattern is visible in the classical Figure 1, where Wilkins calls the extension point F. In the Lean statement the corresponding witness is E. The formal case split generated from this witness is shown separately in Figure 2.

In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
[HilbertIncidence Geo]
(A B C D : Geo.Point) : Prop :=
exists E : Geo.Point,
Geo.Between C B E /\
Geo.AngleCongruent
A B D
A B E
```

Thus the witness E contains exactly the two pieces of geometric information needed by the reconstruction:

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

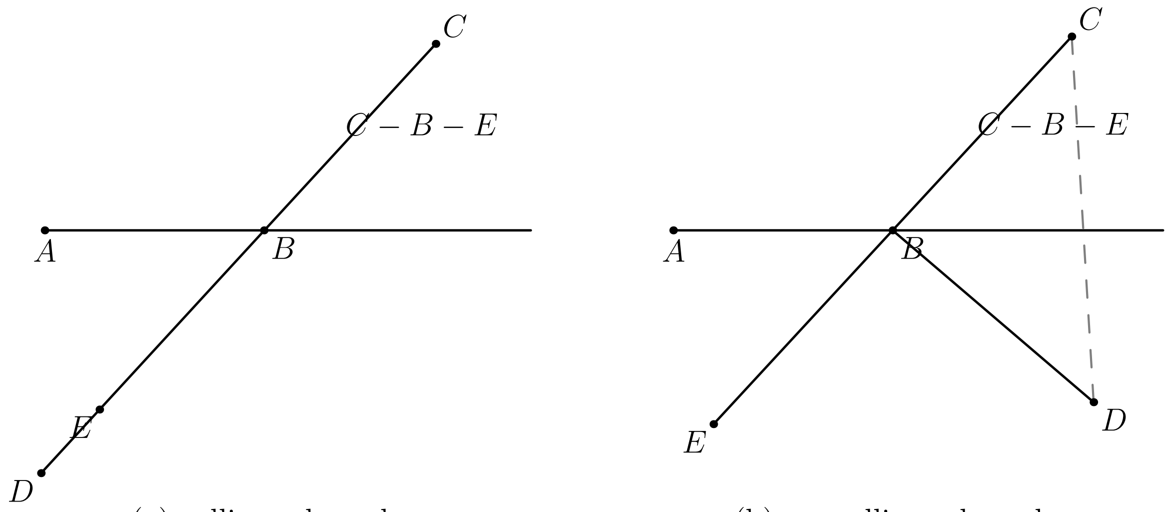
This definition should not be interpreted as a numerical definition of the sum of two angles. It is a synthetic encoding of the particular supplementary configuration occurring in Proposition I.14.

6.2 The Collinear Case

Assume first that C, D, and E are collinear, as in Figure 2(a).

From the definition of `HilbertAnglesEqualTwoRightAngles` we already have

$$C - B - E.$$



(a) collinear branch (b) noncollinear branch

Figure 2: The genuine geometric case split in the current Lean proof. (a) In the collinear branch, C, D, E lie on one carrier and the line-uniqueness argument identifies the crossing with B, yielding $C - B - D$. The displayed placement of D relative to E is only representative; no order between D and E, and not even their distinctness, is assumed. (b) In the noncollinear branch, $C - B - E$ and plane separation give $\mathrm{SameSide}(D, E; \mathrm{base})$; the dashed segment records the opposite-side crossing of CD with the base line.

Hence B lies on the line determined by C and E.

Since C and D lie on opposite sides of the base line AB, the segment CD meets that line. Let X be a point of intersection, so that

$$C - X - D$$

and X lies on the base line.

No separate figure is needed for this branch. The point X is only an incidence witness: if the line through C, D, E meets the base line at X, then line uniqueness forces $X = B$. Drawing X as a second visible point near B is more confusing than informative, because the proof immediately shows that the two points coincide.

Both B and X now lie on two lines: the base line AB and the line containing C, D, E.

Suppose that

$$X \neq B.$$

By uniqueness of the line through two distinct points, the two lines would then have to coincide. In particular, C would lie on the base line.

This contradicts the hypothesis that C and D lie on opposite sides of the base line, since a point belonging to the line itself cannot be one of the two opposite-side points.

Therefore

$$X = B.$$

Substituting this equality into

$$C - X - D$$

gives

$$C - B - D.$$

Thus the collinear branch requires no angle argument at all. Once C, D, E are known to be collinear, the conclusion follows entirely from betweenness, plane separation, incidence, and uniqueness of the line through two distinct points.

6.3 The Noncollinear Case

Assume now that C, D, and E are not collinear.

The segment CD meets the base line because C and D lie on opposite sides of it. The segment CE also meets the base line, namely at B, since

$$C - B - E.$$

Because C, D, E form a genuine triangle, the Hilbert plane-separation theory implies that the remaining endpoints D and E lie on the same side of the base line; this is the configuration shown in Figure 2(b). Thus

$$\mathrm{SameSide}(D, E; AB).$$

We now use Hilbert's angle-construction axiom.

The angle $\angle ABE$ is nondegenerate. Construct, on the side of the base line containing D, a ray BZ satisfying

$$\angle ABZ \cong \angle ABE.$$

Hilbert III.4 includes a uniqueness clause: on the prescribed side of the base ray there is only one ray realizing the prescribed angle. In the current proof this uniqueness closure is the operative output of the construction. The returned facts named `hZDSame` and `hAngleZ` are not used later; the proof retains `hUnique` and applies it twice.

Since E lies on the same side of the base line as D, and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\mathrm{SameRay}(B, Z, E).$$

On the other hand, the encoded hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

and hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

The point D lies on the side on which BZ was constructed. Applying the same uniqueness clause again gives

$$\mathrm{SameRay}(B, Z, D).$$

We therefore have two same-ray relations with the common reference ray BZ:

$$\mathrm{SameRay}(B, Z, E), \qquad \mathrm{SameRay}(B, Z, D).$$

The Book Zero ray calculus now yields

$$\mathrm{SameRay}(B, E, D).$$

Finally, from the encoded supplementary configuration we already know

$$C - B - E.$$

Replacing E by another point D on the same ray from B preserves the betweenness relation. Transport of betweenness along equal rays therefore gives

$$C - B - D.$$

This completes the noncollinear branch and hence the Hilbert reconstruction of Proposition I.14.

7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_14
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase :
HilbertIncidence.OnLine A base)
(hBbase :
HilbertIncidence.OnLine B base)
(hOppCD :
```

```
HilbertOppositeSide Geo C D base)
(hTwoRight :
HilbertAnglesEqualTwoRightAngles
Geo A B C D) :
Geo.Between C B D := by
```

The formal conclusion

$$\text{Between}(C, B, D)$$

expresses directly that B lies between C and D, and hence that the rays BC and BD form one straight line.

The principal dependency structure of the proof is

```
HilbertAnglesEqualTwoRightAngles
        |
        +-- C-B-E
        +-- angle ABD ~= angle ABE
        |
HilbertOppositeSide(C,D; base AB)
        |
case split on Collinear(C,D,E)
        |
   +----+--------------------+
   |                         |
collinear                 noncollinear
   |                         |
line uniqueness           SameSide(D,E; base AB)
   |                         |
X = B                    Hilbert III.4 uniqueness
   |                         |
   |                     same ray BE, BD
   |                         |
   +-----------+-------------+
               |
hilbert_between_transport_sameRays
               |
               v
             C-B-D
```

The proof introduces no new geometric axiom. The new predicate

```
HilbertAnglesEqualTwoRightAngles
```

serves only as an interface representation of the angle hypothesis. The actual derivation uses the existing incidence, order, plane separation, angle-construction, and ray machinery.

A notable difference from Euclid's dependency graph is that the Lean theorem does not invoke Proposition I.13. The supplementary configuration needed in the classical proof has already been encoded in the hypothesis, and the identification of the relevant rays is obtained directly from Hilbert III.4. Likewise Proposition I.4 is historical/foundational background rather than a direct theorem-level dependency of `euclid_proposition_14`.

The current source computes the additional local fact `hOppCE`, asserting that C and E are on opposite sides of the base, but no subsequent expression depends on it. The audited DAG therefore records it as a source-level by-product rather than as a causal step.

8 Relationship with Earlier Propositions

8.1 Proposition I.13

Proposition I.13 gives the direct statement: when one straight line stands on another, the two adjacent angles are either two right angles or together equal to two right angles.

Proposition I.14 is the converse. Starting from the supplementary-angle condition, it proves that the two outer rays form one straight line.

Thus I.13 and I.14 form a natural direct/converse pair.

8.2 Proposition I.4 as Foundational Background

The wider congruence infrastructure has classical antecedents including SAS, but the current `Proposition14.lean` does not call `euclid_proposition_4`. The direct mechanism relevant here is Hilbert III.4, which supplies uniqueness of angle construction on a prescribed side of a ray.

This uniqueness principle plays the role of a rigidity theorem: once two rays on the same side make congruent angles with the same initial ray, they must coincide.

8.3 Transition to Proposition I.15

Propositions I.13 and I.14 establish the supplementary and straight-line geometry surrounding an intersection. Proposition I.15 then turns to the opposite angles produced by two intersecting lines and proves their congruence.

The three propositions therefore form a coherent local block: straight-angle structure, its converse, and vertical angles.

9 Hilbert and Book Zero Components Used

The proof of Proposition I.14 uses several previously established components of the Hilbert and Book Zero layers. Their roles in the reconstruction are recorded here explicitly.

9.1 Collinear branch: incidence and line uniqueness

When C, D, E are collinear, the opposite-side hypothesis supplies a point X with $C - X - D$ on the base line. Since $C - B - E$, the point B also lies on the carrier of C, D, E. If $X \neq B$, the two distinct common points X, B would force that carrier to be the base line by

`HilbertPlaneIncidence.line_unique`, contradicting that C is off the base. Hence $X = B$, and the branch closes as $C - B - D$.

The witness X is not drawn in Figure 2(a): its only purpose is to be identified with the already visible point B.

9.2 Third-side endpoints on the same side

In the noncollinear branch, the points C, D, E form a genuine triangle. The base line meets the interior of CD, because C and D lie on opposite sides of it, and it meets the interior of CE at B, since

$$C - B - E.$$

The theorem

```
hilbert_third_side_endpoints_sameSide
```

then shows that the remaining endpoints D and E lie on the same side of the base line.

Schematically,

$$\begin{array}{c} C - X - D, \\ C - B - E, \\ X, B \in AB \end{array} \qquad \Longrightarrow \qquad \mathrm{SameSide}(D, E; AB).$$

This is the plane-separation step that makes the uniqueness clause of Hilbert's angle-construction axiom applicable to both D and E.

9.3 Hilbert III.4: Uniqueness of Angle Construction

Hilbert's angle-construction axiom provides not only the existence of a copy of a given angle but also uniqueness on a prescribed side of the base ray.

In Proposition I.14 a ray BZ is constructed so that

$$\angle ABZ \cong \angle ABE.$$

Both E and D lie on the prescribed side of the base line. The uniqueness clause can therefore be applied twice, yielding

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D).$$

This is the formal replacement for Euclid's final inference that equal angles on the same side determine the same straight line.

9.4 Book Zero ray composition

The Book Zero theorem

```
bookZero_36_ray3
```

is a ray-composition principle.

In the present proof,

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D)$$

imply

$$\mathrm{SameRay}(B, E, D).$$

Thus the auxiliary ray BZ introduced by Hilbert III.4 can be eliminated, leaving the geometrically relevant statement that BE and BD are the same ray.

9.5 Betweenness transport along same rays

The final step uses

```
hilbert_between_transport_sameRays
```

to transport strict betweenness along corresponding rays.

We know

$$C - B - E$$

and have established

$$\mathrm{SameRay}(B, E, D).$$

The ray from B toward C is unchanged, while E is replaced by the point D on the same ray from B. Therefore the theorem gives

$$C - B - D.$$

This is precisely the formal conclusion of Proposition I.14.

The dependency path of the noncollinear branch can therefore be summarized as

$$
\begin{array}{c}
\text{plane separation} \\
\downarrow \\
\mathrm{SameSide}(D, E; AB) \\
\downarrow \\
\text{Hilbert III.4} \\
\downarrow \\
\mathrm{SameRay}(B, Z, E), \mathrm{SameRay}(B, Z, D) \\
\downarrow \\
\texttt{bookZero\_36\_ray3} \\
\downarrow \\
\mathrm{SameRay}(B, E, D) \\
\downarrow \\
\texttt{hilbert\_between\_transport\_sameRays} \\
\downarrow \\
C - B - D.
\end{array}
$$

10 Formalization Notes

Proposition I.14 is a good example of a classical converse whose formal proof is governed by uniqueness rather than by numerical angle arithmetic.

The hypothesis says that two adjacent angles have the same total effect as a straight angle. In a traditional proof this is often read directly from the diagram. In the Hilbert reconstruction, the decisive step is to compare rays using the uniqueness clause of angle construction.

The proof naturally separates into two cases.

If C, D, E are collinear, the desired straight-line conclusion follows from order, incidence, and line uniqueness.

If they are not collinear, plane separation first puts D and E on the same side of the base. Hilbert III.4 then forces the relevant rays to coincide. The resulting same-ray information is transported back to betweenness, yielding the same conclusion $C - B - D$.

Schematically,

$$
\begin{array}{c}
\text{supplementary-angle hypothesis} \\
\downarrow \\
\text{same-side information} \\
\downarrow \\
\text{angle-construction uniqueness} \\
\downarrow \\
\text{same ray} \\
\downarrow \\
\text{betweenness / collinearity} \\
\downarrow \\
\text{straight line}
\end{array}
$$

This is structurally different from simply writing an equation such as $\alpha + \beta = 180^\circ$. No numerical angle measure is introduced. The straight-line conclusion is recovered entirely from incidence, order, angle congruence, side-of-line information, and uniqueness.

11 Dependency Summary

The current formal dependency structure is branch-sensitive:

$$\begin{array}{c}
\texttt{HilbertAnglesEqualTwoRightAngles} \implies C - B - E,\ \angle ABD \cong \angle ABE \\
\texttt{HilbertOppositeSide}(C, D; \text{base}) \\
\downarrow \\
\text{case split on } \operatorname{Collinear}(C, D, E)
\end{array}$$

$$\begin{array}{cc}
\text{collinear} & \text{noncollinear} \\
\downarrow & \downarrow \\
\text{incidence + line uniqueness} & \texttt{hilbert_third_side_endpoints_sameSide} \\
\downarrow & \downarrow \\
X = B & \text{Hilbert III.4 uniqueness} \\
\downarrow & \downarrow \\
C - B - D & \operatorname{SameRay}(B, E, D) \\
 & \downarrow \\
 & \texttt{hilbert_between_transport_sameRays} \\
 & \downarrow \\
 & C - B - D
\end{array}$$

The source-level fact `hOppCE` is derived before the split but is not consumed by either branch. It is therefore intentionally omitted as a dependency arrow.

12 Diagrammatic Audit

The preserved Book1R chapter contained three geometric floats:

1. the initial configuration with BC and BD on opposite sides of the base;
2. Wilkins's auxiliary extension $C - B - F$;
3. the noncollinear formal branch.

The first float is redundant under the current audit standard. The Wilkins configuration already contains the same initial rays and adds the genuine extension used by the classical proof. It is therefore retained as the sole classical figure.

The current production Lean proof, however, performs a genuine geometric case split on whether C, D, E are collinear. The preserved chapter drew only the noncollinear branch. Figure 2 now records both branches in one two-panel formal figure. The intersection witness X in the collinear branch and the angle-copy witness Z in the noncollinear branch are not drawn separately: X is immediately proved equal to B, while Z exists only to activate Hilbert III.4 uniqueness and is immediately identified at the ray level with both E and D.

Thus the audited diagram set records stable geometric states rather than theorem-call or tactic state.

13 Audit Status

This revision audits the documentation against the currently inspected production source `Proposition14.lean` and the preserved Book1R chapter. The production Lean file has not been modified.

No `lake build` and no fresh `#print axioms` check were run as part of this diagrammatic documentation audit. Consequently this chapter does not claim a new compilation or axiom-audit result.

14 Conclusion

Proposition I.14 completes the converse half of the supplementary-angle theory begun in I.13.

Its formal reconstruction is especially instructive because it avoids introducing angle addition or numerical angle measures. The proof instead uses Hilbert's side-of-line and angle-construction uniqueness machinery to show that the geometry of the rays is rigid enough to force a straight line.

The proposition therefore illustrates a recurring feature of the project: statements that look algebraic in modern notation can often be recovered synthetically from incidence, order, congruence, and uniqueness.

Together with I.13, it prepares the local theory of intersecting lines used immediately in Proposition I.15.